Immobile electrons at frozen gate charge: what the transport data of electrolyte-gated $LaInO_3/BaSnO_3$ allow

Gracen Ownby

University of Oklahoma, Norman, Oklahoma, USA

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Abstract

In the electrolyte-gated $LaInO_3/BaSnO_3$ two-dimensional electron gas of Lee *et al.*, each gate voltage is set above the freezing point of the ionic liquid and the sample is then cooled, so the total charge and the ionic and defect configuration are fixed below it. Yet between 150 and 2 K the Hall density falls by 10, 15, 26 and 40% at 4, 5, 7 and 8 V. We ask what the published Hall, resistance and magnetoresistance data allow, using a self-consistent subband model and a multisubband transport calculation. Coupled subbands in a magnetic field are exactly equivalent to independent Drude channels, so we search all two- and three-channel sets that reproduce the measured conductance and Hall density at 2 K. At 7 V, a magnetoresistance flat to 0.1% between 2 and 7 T then requires about a quarter of the electrons (at least 22% in three channels; 25–29% in two, with mobilities below 50 cm^2/Vs) to sit in states with mobility below 200 cm^2/Vs, against about 1200 cm^2/Vs for the rest. At 5 V the data cannot decide. Coulomb scattering by the gate-induced positive charge next to the interface, which electrostatics requires, makes the subbands differ in mobility by less than a factor of two and lowers the Hall density by at most 4% at 7–8 V, with a magnetoresistance consistent with the data. So the identified scattering produces no such slow channel: at 7 V about 2×10^{13} cm^{-2} electrons are effectively immobile. They are not captured on cooling by traps with levels fixed relative to the band. At fixed charge, taking them out of the band lowers the Fermi level by 0.1–0.3 eV at 7–8 V; a trap level could follow only with an entropy of emptying of more than $10 k_B$, and a linear programme over trap-level spectra, freezing temperatures and entropy factors finds no physical trap set at 7–8 V with fewer states than the interface has unit cells, and at 5 V it needs nearly one per unit cell. A fixed slow fraction with changing mobility can be fitted at every temperature, but it would have to conduct like a metal at 10 K and stop by 2 K, and at 8 V it predicts a 5% Hall nonlinearity at 50–100 K, whereas Lee *et al.* report linear Hall resistances at all temperatures; how the immobile fraction forms remains open. Below 50 K the conductance falls with the Hall density by 9–24% at 5–8 V, while the field-suppressible weak-localization correction is only 2–4%. The conductance also bounds how close random point charges can be to the interface; at 8 V the bound and the electrostatic limit agree within a factor of about 1.5 for clean $BaSnO_3$ and singly charged centres.

1 Introduction

The two-dimensional electron gas (2DEG) at the polar interface between $LaInO_3$ and $BaSnO_3$ combines a polarization-induced density of a few 10^{13} cm^{-2} with the high mobility of $BaSnO_3$ [1–3]. Lee *et al.* gated it

with the ionic liquid DEME-TFSI on a 5 nm $LaInO_3$ cap, grown by pulsed laser deposition on 110 nm of $BaSnO_3$ grown by molecular-beam epitaxy on MgO [4]. Each gate voltage V_{GS} was applied at 220 K, where the liquid is mobile; the sample was then cooled to 2 K at constant bias and measured while warming. The liquid has its glass transition near 182 K [5]. Below it the ions do not move, and neither does the positive charge that balances the gated electrons. A companion study [6] found that this positive charge must sit within about one $LaInO_3$ unit cell of the interface to keep electrons out of the cap. It cannot be the layer of liquid ions on the cap surface, 5 nm away; it must be charged defects formed electrochemically during gating, such as oxygen vacancies, whose number is set at 220 K. Lee *et al.* also report an irreversible change of the sheet resistance after gating, consistent with such electrochemistry.

Their Fig. 3 gives the sheet resistance from 2 to 220 K and the Hall density n_H and mobility μ_H at 2, 10, 50, 100 and 150 K for $V_{GS} = 0$–8 V; their Fig. 5a gives the magnetoresistance at 2 K up to 7 T for 0, 1, 3, 5 and 7 V. At 8 V the Hall density reaches 1.08×10^{14} cm^{-2} at 150 K and the Hall mobility 2000 cm^2/Vs at 2 K. Lee *et al.* note that the Hall density falls on cooling at high bias, a point they leave open, and attribute a slow rise of the resistance at low temperature to weak localization.

With the total charge fixed, electrons cannot disappear. Either the Hall measurement undercounts them, or some of them stop contributing to the current. For parallel channels, $n_H = (\sum_i n_i\mu_i)^2/\sum_i n_i\mu_i^2$ equals the electron density only when all channels have the same mobility [7], and at 10^{14} cm^{-2} the interface has two or three occupied subbands. The mobility differences that lower n_H, however, also produce a positive classical magnetoresistance. Here we use this link, with a self-consistent subband model and a multisubband transport calculation, to find what kind of electron distribution the data require. We also use the measured conductance to bound where the gate-induced charge can sit.

2 Data

We digitized three figures of Ref. [4]: Fig. 3b (Hall density and mobility); Fig. 3a (sheet resistance) for 5–8 V, where the curves are separated; and Fig. 5a (magnetoresistance at 2 K) for 3, 5 and 7 V. We read Fig. 3b twice, by hand and by the centroids of the markers, and checked both against their Fig. 4, which plots the Hall mobility against the Hall density at 10 and 150 K. At 4–8 V the readings agree with Fig. 4 to within 0.6% in density and 1.5% in mobility; at 5 V the three readings of the density at 2–150 K agree to within 1%. At 2 K and 8 V we read 6.50×10^{13} cm^{-2} and 2010 cm^2/Vs, against 6.52×10^{13} cm^{-2} and 1999 cm^2/Vs quoted in the text of Ref. [4]. The 4 V mobility marker at 10 K is hidden in Fig. 3b, and we take it from Fig. 4. The 6 V markers at 2 K are hidden behind the 7 V markers, so we do not use the 6 V Hall data. The magnetoresistance is read to about 0.05 percentage point.

For parallel Drude channels the conductance is $\sigma = e\,n_H\mu_H$ exactly. The resistance of Fig. 3a is 1.12–1.48 times $1/(e\,n_H\mu_H)$, a factor that grows with gate voltage and changes with temperature. Both were measured on the same Hall bar, so the difference is not geometric, and we cannot explain it. Their relative temperature dependence agrees: between 50 and 2 K the resistance rises by 9.7, 18.5 and 26.8% at 5, 7 and 8 V in Fig. 3a and by 9.7, 24.5 and 31.1% in the Hall data (Fig. 1b), so the conductance falls by 9–24%. The rise is not monotonic in gate voltage. At 6 V the resistance of Fig. 3a falls by 10% between 50 and 2 K, and at 4 V the Hall data give a fall of 39% with a Hall mobility rising by 76%.

Since $n_H \le n_{\rm tot}$, the largest measured Hall density, at 150 K, is a lower bound on the electron density $n_{\rm tot}$; we use $n_{\rm tot} = n_H(150\ {\rm K})$. At 2 K the Hall density is 90, 85, 74 and 60% of it at 4, 5, 7 and 8 V (Fig. 1a), a

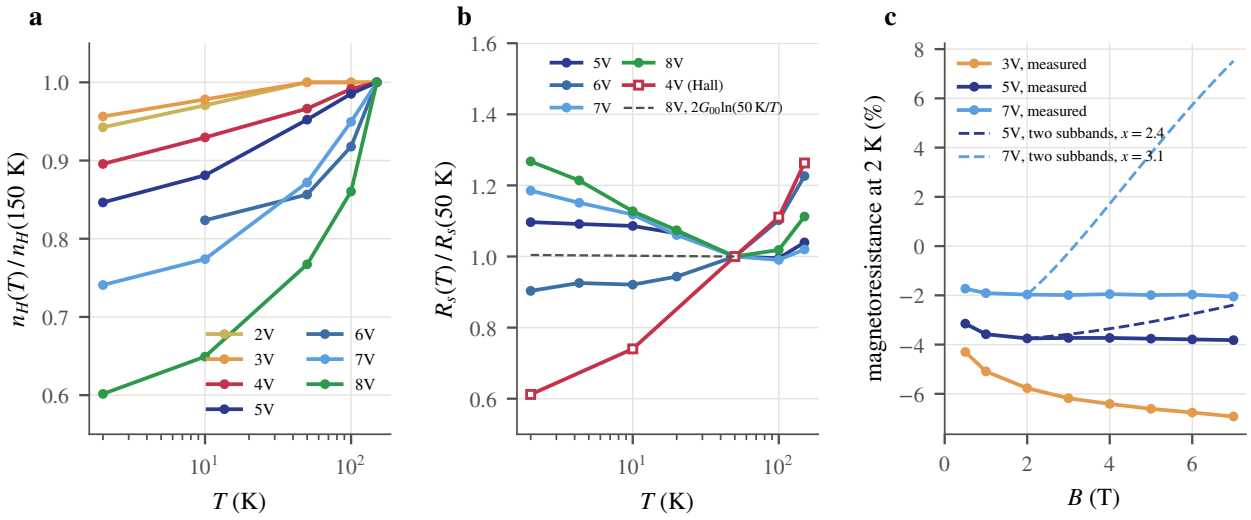


Figure 1: Data of Lee *et al.* [4], digitized. (a) Hall density relative to its 150 K value, with the total charge frozen. (b) Sheet resistance relative to 50 K, from their Fig. 3a (5–8 V) and from $1/(e n_H \mu_H)$ (4 V). Dashed: the 8 V curve if the conductance fell by $2G_{00}\ln(50\,\mathrm{K}/T)$, a typical size for weak localization and the interaction correction together. (c) Magnetoresistance at 2 K (their Fig. 5a), mean of $\pm B$. Dashed: the classical magnetoresistance, added to the measured value at 2 T, of two independent subband channels with the model's populations, the measured conductance and the mobility ratio x that reproduces the 2 K Hall density at low field with the upper subbands more mobile.

shortfall of 0.4, 0.7, 1.9 and 4.3×10^{13} cm^{-2}. At 2–3 V the sheet resistance is 2.5–3 kΩ ($k_F l \approx 10$), close to the metal–insulator crossover, so we concentrate on 4–8 V.

3 Model

Subbands. The interface is described by the self-consistent Schrödinger–Poisson model of Refs. [6, 8], implemented in qwkit [9]. Its ingredients are:

- a Kane-nonparabolic conduction band of $BaSnO_3$ (mass $0.19\,m_0$, nonparabolicity 0.56 eV^{-1}, static permittivity 20);
- a 5 nm $LaInO_3$ cap (offset 1.6 eV, mass $0.46\,m_0$);
- deep acceptors in the $BaSnO_3$, depleting over a width set by Fermi-level pinning;
- a fixed interface charge that gives 5×10^{12} cm^{-2} without gating.

We consider strongly compensated $BaSnO_3$ ($N_A = 5 \times 10^{19}$ cm^{-3}, as reported for films grown by pulsed laser deposition [2]) and clean $BaSnO_3$ (10^{18} cm^{-3}). The added positive charge sits next to the interface, as Ref. [6] requires; by Gauss's law the $BaSnO_3$ side does not depend on its exact place while the cap is empty. Ten subbands enter at 2 K on a 0.5 Å grid. In compensated $BaSnO_3$ the second subband is occupied only above 6.5×10^{13} cm^{-2} and holds 3 and 12% of the electrons at 7 and 8 V. In clean $BaSnO_3$ the ground subband holds 84, 81, 79 and 76% at 4, 5, 7 and 8 V. Its mean distance from the interface is 0.7–1.1 nm, against 2.0–3.0 nm for the second subband.

Scattering. At 2 K we compute the momentum relaxation times of all occupied subbands at the Fermi level from the coupled Boltzmann equations of Siggia and Kwok [10],

$$1 = \sum_j \left[\frac{\tau_i}{\tau_{ij}^{(0)}} - \frac{v_j}{v_i} \frac{\tau_j}{\tau_{ij}^{(1)}} \right], \tag{1}$$

with Fermi velocities $v_i = \hbar k_i / m_i$ and nonparabolic density-of-states masses m_i. The matrix elements are those of point charges in the Born approximation, screened in the static random-phase approximation with the intrasubband polarizabilities of all occupied subbands [10, 11]. The scatterers are:

1. all of the positive charge added by gating, $n_{\mathrm{tot}} + N_{\mathrm{dep}} - \sigma_P$, as random point charges Ze ($Z = 1$, 2) in a plane a distance $d = 1$–32 Å inside the cap. They act through the cap–$BaSnO_3$ boundary, with cap permittivity $\varepsilon_{\mathrm{cap}} = 24$, 38 or 50: the bare potential in the $BaSnO_3$ is reduced by $2\varepsilon/(\varepsilon + \varepsilon_{\mathrm{cap}})$, and the 2DEG's own screening acquires an image term;

2. the ionized acceptors of the depletion layer;

3. charged threading dislocations at 10^{10}–10^{11} cm^{-2}, typical of $BaSnO_3$ grown by molecular-beam epitaxy [12], whose bare potential acts only within a subband [13];

4. interface roughness in the finite-barrier form [11], as an order-of-magnitude check.

The relaxation times agree with an independent quadrature of the single-subband integral to 10^{-3}, and with a direct angular solution of the coupled Boltzmann equations to 10^{-9}. The mobilities change by less than 0.2% between 32 and 128 angles and on a 0.25 Å grid, and by 0.7–1.5% between 10 and 14 subbands.

Magnetotransport. For isotropic subbands the drift velocity u_i of each subband obeys

$$m_i \sum_j G_{ij} u_j = -e(E + u_i \times B), \tag{2}$$

with G the relaxation matrix of the coupled Boltzmann equations. Detailed balance makes G similar to a symmetric matrix, so the subbands can be replaced exactly by independent Drude channels, the eigenmodes of $m_i G_{ij}$, with non-negative weights that sum to n_{tot}; we verified this numerically to 10^{-9}. Intersubband scattering therefore changes the weights and mobilities of the channels, which can suppress or enhance the magnetoresistance relative to the subbands taken as independent [14], but it cannot produce a Hall response outside what some set of independent channels gives. Lee *et al.* obtained n_H from Hall resistances that were linear up to 7 T; we fit a straight line through the origin over 0.25–7 T to model the same procedure.

4 Results

4.1 What the magnetoresistance requires

At 2 K the magnetoresistance at 5 and 7 V (Fig. 1c) has a weak-localization cusp below 1 T and is then flat: between 2 and 7 T it changes by -0.07 to -0.12 (5 V) and -0.06 to -0.08 (7 V) percentage points, depending on the reading; we use -0.07 and -0.08. If the upper subbands were the more mobile ones, the 2 K Hall density would need a mobility ratio $x = 2.4$ (5 V) or 3.1 (7 V). Two independent subband channels

with the model's populations and the measured conductance would then raise the magnetoresistance between 2 and 7 T by 1.3 and 9.5 points (dashed in Fig. 1c).

The general question is which channel sets reproduce the measured conductance and fitted Hall density while keeping the rise between 2 and 7 T within the measured change plus a tolerance. We take tolerances of 0.5 and 1.0 point. They cover the digitization, a remaining weak-localization tail (about -0.2 and -0.05 points at 5 and 7 V, scaled from the 3 V curve by the conductance), and the parabolic negative magnetoresistance of the electron–electron interaction (a few tenths of a point at 7 V). We sampled two- and three-channel sets with random weights and mobilities between 1 and 30 000 cm^2/Vs, constrained to the measured values (Table 1, Fig. 2b).

- *7 V.* Every acceptable set has a slow channel. With two channels, 25–29% of the electrons must have a mobility below 23 cm^2/Vs (tolerance 0.5) or 49 cm^2/Vs (tolerance 1.0). With three channels, at least 22–23% must be slower than 200 cm^2/Vs. The other electrons then carry the current with about 1200 cm^2/Vs. Sets with a fast minority channel either bend the magnetoresistance or, when that channel saturates below 2 T, bring the fitted Hall density up to n_{tot}.
- *5 V.* The magnetoresistance does not constrain the sets ($\mu B \approx 0.3$ at 7 T): acceptable sets include fast-minority, slow-minority and slow-majority channels, with no slow channel required.

A slow channel holding a quarter of the electrons at 7 V could be a subband 25–50 times less mobile than the others, or electrons in localized states; the transport data alone cannot tell which. Paramagnetic or Kondo-like negative magnetoresistance from spin-carrying centres could in principle cancel a positive classical rise. Doing so at both 5 and 7 V over 2–7 T would need fine tuning.

The magnetoresistance also measures weak localization. The negative cusp saturates at -3.8% (5 V) and -2.0% (7 V), so the whole field-suppressible correction to the 2 K conductance is 2–4%. The resistance, by contrast, rises between 50 and 2 K by 10% at 5 V and 19–24% at 7 V. We write the conductance change as

$$\sigma(50\,\mathrm{K}) - \sigma(2\,\mathrm{K}) = A\,G_{00}\ln 25, \qquad G_{00} = \frac{e^2}{2\pi^2\hbar}, \tag{3}$$

and find $A = 5$–7 at 5 V, 37–66 at 7 V and 98–164 at 8 V, with $k_F l = 56$–540. Weak localization and the electron–electron interaction correction together give A of a few, at most about 4–6 with several subbands and strong dephasing [15, 16]. Their relative size also falls as $1/k_F l$, whereas here it grows with $k_F l$. The resistance rise at 7–8 V is therefore not a quantum correction; at 5 V the comparison is inconclusive. Weak localization leaves the Hall coefficient unchanged, and the interaction correction changes it by twice the relative conductance change [15, 16], which would make the Hall mobility rise as the conductance falls. Between 50 and 2 K, the Hall mobility instead changes by $+3$, -5 and -3% at 5, 7 and 8 V, while the Hall density falls by 11, 15 and 22%.

4.2 What the gate-induced charge does

Electrostatics places the positive charge that balances the gated electrons within a few ångströms of the interface [6], and at these densities it dominates the elastic scattering that we can identify. With the charge 1–6 Å inside the cap ($\varepsilon_{\mathrm{cap}} = 24$), it makes the upper subbands more mobile than the ground subband, which overlaps it most, by a factor of:

- 1.47–1.72 at 4 V;

Table 1: Clean $BaSnO_3$, by gate voltage. Columns: electron density $n_{tot} = n_H(150\,K)$ (10^{13} cm^{-2}); measured n_H/n_{tot} at 2 K; ground-subband fraction f_1; smallest weight of channels slower than 200 cm^2/Vs among the three-channel sets that reproduce the 2 K conductance, Hall density and flat magnetoresistance (tolerance 0.5 / 1.0 point; magnetoresistance measured at 5 and 7 V only); n_H/n_{tot} from Coulomb scattering by the gate-induced charge at the measured conductance (both limits, $Z = 1$, 2, $\varepsilon_{cap} = 24$); smallest distance d_{min} (Å) of that charge compatible with the measured conductance, from the Hall data / from Fig. 3a of Ref. [4], for $Z = 1$ and $Z = 2$; largest distance x_{max} (Å) allowed by electrostatics at 150 K for $\varepsilon_{cap} = 24$ [6].

V_{GS} (V)	n_{tot}	n_H/n_{tot}	f_1	slow weight	n_H/n_{tot}, Coulomb	d_{min}, $Z = 1$	d_{min}, $Z = 2$	x_{max}
4	3.77	0.90	0.84	–	–	≤ 1 / –	≤ 1 / –	39.4
5	4.90	0.85	0.81	0.07 / 0.00	0.92–1.00	≤ 1 / ≤ 1	≤ 1 / ≤ 1	24.1
7	7.33	0.74	0.79	0.23 / 0.22	0.97–0.99	2.8 / 1.6	5.6 / 4.2	11.5
8	10.80	0.60	0.76	–	0.96–1.00	5.0 / 3.5	7.8 / 6.0	4.2

- 1.67–1.86 at 5 V;
- 1.32–1.57 at 7 V;
- 1.43–1.76 at 8 V.

The ratio falls towards 1, and below, as the charge moves further away. The model mobilities are higher than the measured ones, so part of the scattering is unidentified: 70–95% of it at 5 V, less at 7–8 V. We add it in two limits that bracket its effect: all rates scaled by one factor, which keeps the model's mobility ratios, or an extra rate equal for all subbands, which pushes them towards 1. In both, the conductance is matched to the measured 2 K value; we keep only distances at which the model conductance is not already below it. With the coupled magnetotransport, the Coulomb scattering then lowers the Hall density to 0.92–1.00 (5 V), 0.97–0.99 (7 V) and 0.96–1.00 (8 V) of n_{tot} (Fig. 2a), with a magnetoresistance rise of at most 0.3, 0.7 and 1.4 points between 2 and 7 T. That is a deficit of 0–8% at 5 V, against 15% observed, and at most 3–4% at 7–8 V, against 26–40%.

Charged dislocations scatter most strongly the subbands with the smallest Fermi wavevector and lower the ratio, to 0.33–1.27 at 4 V and 1.00–1.43 at 8 V for 10^{10}–10^{11} cm^{-2}. Interface roughness leaves it at 1.3–1.9. In compensated $BaSnO_3$ the barely occupied second subband scatters more strongly than the ground subband ($x = 0.1$–0.9), but holds too few electrons (3–12%) to form the slow channel that 7 V requires. None of the scatterers we can identify makes a quarter of the electrons 25–50 times slower than the rest.

4.3 Where the gate-induced charge can be

The conductance limits how strongly the gate-induced charge can scatter. Suppose other scattering only adds to it and the subband populations do not change with temperature. Then the conductivity mobility limited by that charge alone, $\mu_\sigma = \sum_i (n_i/n_{tot})\mu_i$, cannot be smaller than the largest measured $\sigma/(en_{tot})$ between 2 and 150 K. From the Hall data this bound is 489, 392, 1128 and 1585 cm^2/Vs at 4, 5, 7 and 8 V; from the resistance of Fig. 3a it is 303, 812 and 1066 cm^2/Vs at 5, 7 and 8 V. If a quarter of the electrons are immobile, the bound on the rest is higher still (but see the limitations in Sec. 5).

Figure 2c shows μ_σ against the distance of the charge at 8 V, with the permittivity entering the scattering and the electrostatic limit consistently:

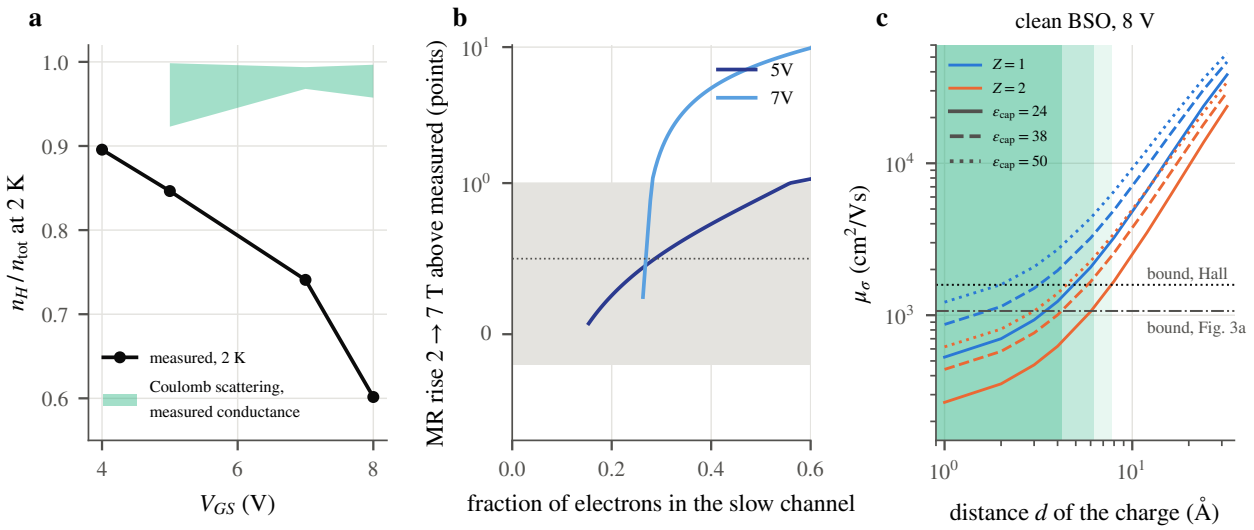


Figure 2: (a) Hall density relative to n_{tot} at 2 K: measured, and the range produced by Coulomb scattering by the gate-induced charge at the measured conductance (both limits for the unidentified scattering; $Z = 1$, 2; $\varepsilon_{cap} = 24$). (b) Two channels that reproduce the measured 2 K conductance and fitted Hall density, as a function of the fraction of electrons in the slower one: rise of the classical magnetoresistance between 2 and 7 T above the measured change. Shaded: within a 1.0-point tolerance; dotted: 0.5 point. At 7 V only a slow fraction of 0.26–0.29 is acceptable. (c) Conductivity mobility limited by the gate-induced charge, as random point charges Ze a distance d inside the cap, at 8 V for clean $BaSnO_3$ and cap permittivities 24, 38 and 50. Horizontal lines: the lower bounds from the measured conductance (Hall data and Fig. 3a of Ref. [4]). Shaded: distances allowed by electrostatics at 150 K [6] (darkest: $\varepsilon_{cap} = 24$).

- At 4–5 V the bound is met even at 1 Å.
- At 7 V it requires $d \geq 2.8$ or 1.6 Å for $Z = 1$ (Hall or Fig. 3a conductance) and $d \geq 5.6$ or 4.2 Å for $Z = 2$, well inside the electrostatic limit of 11.5 Å.
- At 8 V it requires $d \geq 5.0$ or 3.5 Å for $Z = 1$ and 7.8 or 6.0 Å for $Z = 2$. Electrostatics allows 4.2 Å for $\varepsilon_{cap} = 24$, and 6.2 and 7.7 Å for 38 and 50.

For singly charged centres, the two limits meet within a factor of 1.2 in the scattering rate at the measured cap permittivity. Doubly charged centres would need their rate reduced 1.6–2.4 times, or a cap permittivity of at least 38. Spatial correlation of the charged centres reduces their scattering in this way, strongly so at high density [17, 18]. These numbers come from an envelope-function model at the edge of its validity for charges a few ångströms from the interface, and the electrostatic limit treats the charge as a smeared sheet, so we read them as consistent within a factor of about 1.5.

For strongly compensated $BaSnO_3$, in contrast, the conductance at 8 V requires $d \geq 6.5$–12.4 Å for $\varepsilon_{cap} = 24$, while electrostatics allows 1.7 Å, so the scattering rate would have to be reduced 3–8 times. For $\varepsilon_{cap} = 50$, $d \geq 3.5$–8.5 Å against 2.8 Å, a factor of 1.2–3.0. Roughness with correlation lengths of 3 nm or less would by itself bring the conductivity mobility below the 7–8 V bounds. At 6–20 nm it lowers μ_σ by 3–19% for 2 Å steps and by 9–48% for 4 Å steps.

5 Discussion

A quarter of the electrons immobile at 7 V. The magnetoresistance requires about a quarter of the electrons at 7 V to be nearly immobile, and the scattering we can identify gives no such channel. We therefore read the Hall-density drop at 7 V as electrons that no longer carry current: 25–29% of 7.3×10^{13} cm^{-2}, about 2×10^{13} cm^{-2}, immobile at 2 K at fixed total charge. Below 50 K the conductance falls with the Hall density at nearly constant Hall mobility, as expected if the remaining electrons keep their mobility. At 8 V there is no magnetoresistance; there the Hall linearity reported by Lee *et al.* argues against a fast minority channel, which would bend the Hall resistance by 9% over 0–7 T, but not against an immobile fraction, which leaves it straight. At 5 V the data allow either reading.

Inhomogeneity. A film in which some regions stop conducting does not mimic this. In two dimensions a conducting film with insulating holes has the Hall coefficient of the conducting part, for weak fields and non-percolating holes [19], so regions that stop conducting raise the resistance without lowering the Hall density; in two dimensions a uniform density with any spatial variation of the conductivity gives $R_H = 1/(ne)$ at weak field. Regions of different density that all conduct behave, in a parallel geometry, as channels of different weight and mobility, and the analysis of Sec. 4.1 applies to them; for random two-dimensional mixtures the Hall response is more complex and we do not exclude them rigorously.

The electrons are not captured on cooling. At fixed total charge, electrons that leave the band for traps or other localized states take the Fermi level down with the free density. In the model, the fall of the Hall density between 150 and 2 K lowers E_F by 40–52, 91–103 and 171–182 meV relative to the lowest subband at 5, 7 and 8 V (compensated and clean $BaSnO_3$), and by 77–79, 159–170 and 283–333 meV relative to the $BaSnO_3$ band edge at the interface (Fig. 3a). A trap level fixed relative to the band fills on cooling only if its free energy of emptying, $E_t + TS_t$ relative to E_F, with $S_t = S_{\mathrm{empty}} - S_{\mathrm{filled}}$ the entropy gained on emptying, rises with temperature at least as fast as E_F does, which requires $S_t \geq \Delta E_F/\Delta T$. Between 2 and 10 K, where the Hall density rises by 4–8% at 5–8 V, E_F rises by 10–24 meV relative to the lowest subband and 18–45 meV relative to the interface band edge, so S_t would have to be 14–35 k_B or 26–65 k_B, a degeneracy ratio between empty and filled trap of at least $e^{14} \approx 10^6$. This interval is only about twice the digitization tolerance, but at 7–8 V the requirement is also 10–16 k_B (lowest subband) and 19–30 k_B (interface band edge) over 10–50 K. At these temperatures vibrational and excited-state entropies vanish, and the ground-state degeneracies of point defects are a few. Slow kinetics does not remove the problem. A trap whose capture and emission freeze at T_f during cooling keeps the occupation it had at T_f until it is warmed past T_f again, so each trap follows its equilibrium occupation at $\max(T, T_f)$ along the cooling curve; freezing can hold occupations but cannot reverse the sign of their change.

A linear programme makes this quantitative without assuming a trap spectrum. It finds the smallest number of trap states that reproduces the measured $n_H(T)$ at all temperatures to within 2%, allowing any distribution of trap levels, any distribution of freezing temperatures, any total charge above $n_H(150\,\mathrm{K})$, trap levels tied to the lowest subband, to the interface band edge or to the bulk, and a constant entropy of emptying S_t from -5 to $40\,k_B$. For $|S_t| \leq 2\,k_B$, the fewest trap states are 5×10^{12} (4 V), 5×10^{14} (5 V), 1.5×10^{16} (7 V) and more than 10^{18} cm^{-2} (8 V), against 5.9×10^{14} cm^{-2} unit cells per interface plane. The minimum falls to the deficit itself only for $S_t \gtrsim 10\,k_B$ at 7 V and $S_t \gtrsim 20\,k_B$ at 8 V, for the most favourable reference. Only the 10% fall at 4 V, within the 2% tolerance, can be reproduced with physical trap parameters.

What a fixed slow fraction would require. If the electrons are not exchanged, the populations stay fixed and only their mobilities change. Two parallel channels with fixed densities, $n_f = n_H(2\,\mathrm{K})$ and $n_s = n_{\mathrm{tot}} - n_f$,

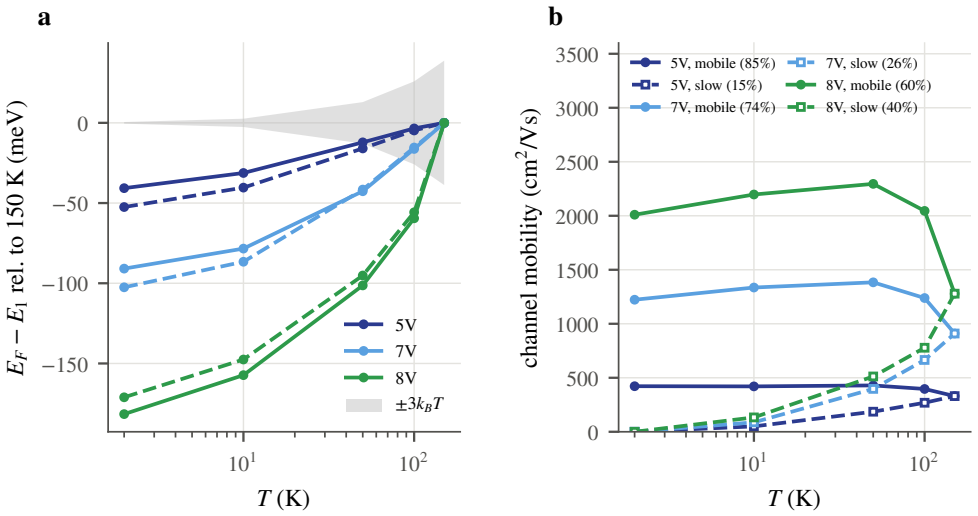


Figure 3: (a) Fermi level relative to the lowest subband, relative to its 150 K value, if the free density were the measured Hall density at fixed total charge (solid: clean, dashed: compensated $BaSnO_3$), with $\pm 3k_BT$ shaded. (b) Two parallel channels with fixed densities ($n_f = n_H(2\,\mathrm{K})$, $n_s = n_{\mathrm{tot}} - n_f$) whose mobilities reproduce the measured conductance and Hall density at every temperature (a reparametrization): mobile (filled) and slow (open) channel.

reproduce any measured conductance and Hall density, two equations for two mobilities at each temperature, so this is a reparametrization rather than a test; but it makes the requirements concrete (Fig. 3b). At 7 V the mobile channel has 1223, 1336, 1384, 1238 and 910 cm^2/Vs at 2, 10, 50, 100 and 150 K, and the slow channel, 26% of the electrons, 0, 87, 397, 665 and 910 cm^2/Vs. Its sheet conductance is already 7 and 31 e^2/h at 10 and 50 K (24 and 91 e^2/h at 8 V), far above what hopping can carry, and vanishes at 2 K: the slow electrons would have to conduct like a metal at 10 K and stop conducting by 2 K. The same channels predict, at 8 V, a Hall resistance at 7 T 4.9 and 4.8% below its low-field line at 50 and 100 K and a classical magnetoresistance of 8–14% at 7 T between 10 and 100 K (at 7 V, 1.3% and 3.5% at 50 K). Lee *et al.* report Hall resistances linear up to 7 T at all temperatures and gate voltages [4], which is in tension with the 8 V predictions. With n_{tot} 5–10% above the 150 K Hall density, the slow fraction at 7 V is 29–33% and the requirements are similar.

What remains open. The 2 K data require about a quarter of the electrons at 7 V to be nearly immobile. How that fraction forms on cooling at fixed charge is not explained by capture into traps, which thermodynamics excludes at 7–8 V for trap levels fixed relative to the band and makes implausible at 5 V, and only uneasily by a fixed slow fraction, which needs a metal-like channel that switches off below 10 K and a Hall nonlinearity at 8 V that is in tension with the reported linearity. The cap itself remains empty: at 8 V its lowest states are 0.23 eV above the Fermi level for clean $BaSnO_3$ [6]. Paramagnetic centres would contribute a negative magnetoresistance at 2 K; the flat magnetoresistance argues against strong spin-flip scattering. The non-monotonic gate dependence of the resistance rise (none at 4 and 6 V) points to a process that depends on the sample or the cool-down.

What the mobility means. If the deficit is immobile electrons, the Hall mobility is close to the mobility of the conducting ones: 2010 cm^2/Vs at 8 V and 2 K, for about 60% of the gate-induced electrons. The conductance per gate-induced electron, $\sigma/(en_{\mathrm{tot}})$, is 1210 cm^2/Vs at 2 K and 1590 cm^2/Vs at 50 K.

Limitations.

- The analysis rests on digitized data from one publication. Its Supporting Information, which holds the Hall resistances, was not available to us.
- The magnetoresistance was shown up to 7 V only, so the slow-channel requirement is direct at 7 V and not tested at 8 V.
- The total density is taken as the 150 K Hall density, a lower bound; a larger one would increase the immobile fraction.
- The trap analysis uses the model's Fermi level as a function of the free density and treats trap levels as fixed relative to one of three references, with a constant entropy factor; it assumes equal cooling and warming curves. Traps whose levels depend on the local field or on their own occupation are not included. The linear programme becomes numerically unreliable above about 10^{18} cm^{-2}, where we report only that the number exceeds the interface sites.
- The two-channel description of a fixed slow fraction is a parametrization; the reported Hall linearity is qualitative, and the Hall resistances themselves (Supporting Information of Ref. [4]) were not available.
- The conductance bound of Sec. 4.3 places all n_{tot} electrons in the 2DEG. If a quarter of them form a separate immobile subsystem, both the bound and the model mobility change, and we have not redone the comparison for that case.
- The transport model uses the Born approximation and static screening, treats the gate-induced charge as uncorrelated point charges in a plane, and neglects exchange–correlation and phonons. Most of the scattering at 5 V is not identified.
- The roughness matrix element is first order in the step height and omits the mass-discontinuity term, so roughness is an order-of-magnitude check only.

6 Conclusion

In the electrolyte-gated $LaInO_3/BaSnO_3$ 2DEG of Lee *et al.*, the Hall density at fixed total charge falls by up to 40% between 150 and 2 K. Coupled subbands are equivalent to independent channels, and at 7 V the published magnetoresistance, flat between 2 and 7 T, then requires about a quarter of the electrons to be nearly immobile. Coulomb scattering by the gate-induced charge, the dominant identified scatterer, changes the Hall density by only a few per cent. Capture on cooling into traps with levels fixed relative to the band cannot account for the drop: at 7–8 V the Fermi level would have to fall by 0.1–0.3 eV, which a trap can follow only with an unphysical entropy, and no physical trap spectrum reproduces the data at 7–8 V with fewer states than the interface has unit cells (at 5 V, nearly one per unit cell). A fixed slow fraction in parallel with the 2DEG can be fitted, but it needs a metal-like channel that stops conducting below 10 K and predicts a Hall nonlinearity at 8 V that is in tension with the reported linearity, so the origin of the immobile electrons remains open. The rise of the resistance at 7–8 V is far too large for weak localization, which the magnetoresistance limits to 2–4%. Together, the conductance and the electrostatic limit place the gate-induced charge within a few ångströms of the interface for clean $BaSnO_3$, consistent within a factor of about 1.5 if it consists of singly charged or correlated centres, and they argue against strongly compensated $BaSnO_3$. The calculations use qwkit [9].

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